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RESEARCH ARTICLE

Three-Way Clustering Based on Digital Image Processing

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ABSTRACT A three-way clustering algorithm based on image processing is proposed by combining blurring and sharpening operations in digital image processing. The proposed algorithm quantifies the density of the dataset into gray values through inverse multivariate quadratic kernel function. By blurring and sharpening operations for the university, we delete the samples with low density and obtain samples with high density. Then, different clusters are produced by traditional clustering algorithm for samples with high density. For each cluster, the core region and fringe region are obtained through blurring operation and sharpening operation, respectively. The experiments on different UCI datasets verify that the clustering algorithm is stable and effective by comparing the clustering indexes adjusted rand index (ARI), normalized mutual information (NMI) and adjusted mutual information (AMI).

INDEX TERMS Three-way decision, three-way clustering, kernel density estimation, digital image processing.

I. INTRODUCTION

As an efficient technology in data mining, clustering has been widely used in various areas [1], including information granulation [2], information fusion [3], attribute reduction [4], etc. Specifically, its objective is to find multiple sets of objects that are similar to one another in the same cluster but dissimilar to objects in any other clusters [5]. This allows for the discovery of the inherent data organizational structure within objects of unknown categories. There exist many varieties of clustering algorithms [6], [7], which can be roughly categorized into hard clustering and soft clustering. Hard clustering exactly assigns each object to one cluster with a crisp boundary [8], which is suitable for datasets with clear structures. Differently, soft clustering allows samples to belong to multiple clusters with a certain probability or membership degree, providing a more flexible clustering strategy [9].

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In recent years, a theory of three-way decisions was proposed by Yao [10] to address the problem of information uncertainty in data science. Three-way decision expands the decision-making results to acceptance, rejection, and delayed decisions, which becomes a paradigm of information processing based on triadic patterns [11], [12]. The triadic thinking of three-way decision offers a novel motivation and strategy for clustering data with uncertainties, leading to the rise of three-way clustering [13]. By applying three-way decisions in the field of clustering, Yu [14] introduced three-way clustering to capture the uncertainty of clustering by adding a fringe region for uncertain samples. Three-way clustering divides the data universe into three disjoint parts [15]: the core region (Co), the fringe region (Fr), and the trivial region (Tr). The Co defines the inside region and corresponds to the highly concentrated objects of a cluster. The Fr is an area where the objects are loosely concentrated, and there may be common objects in the fringe regions of different clusters. The Tr can be expressed as the complement of the union of Co and Fr. Fig. 1 illustrates the comparison of representation results between three-way

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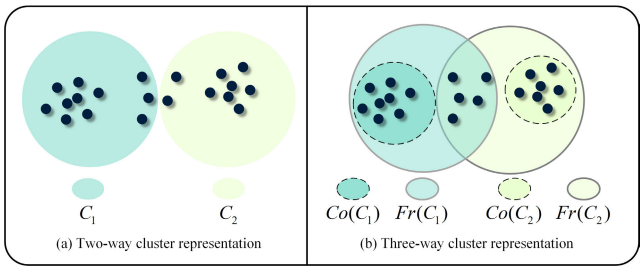


FIGURE 1. Cluster representations of two-way clustering (a) and three-way clustering (b).

clustering and two-way clustering [16]. As shown in Fig. 1(b), three-way clustering with interval sets can distinguish the intension and extension of each concept, contributing to better processing the data in the overlapping region. Under the framework of three-way clustering, some three-way clustering algorithms [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27] have been proposed in recent years.

Though many efficient approaches have been proposed and studied, most approaches predominantly apply a pair of partition thresholds across the entire datasets to define the three regions [23]. The algorithms with fixed and global thresholds disregard the clusters with different distributions and densities [28]. For instance, Yu presents the TWD-CI algorithm for incomplete data, which divide original incomplete data set into four parts depending on the parameters [14]. Chu et al. proposes a neighborhood rough set-based three-way clustering on the hybrid information system, which employs an evaluation function with pairs of thresholds to quantify the relationship between an object and a cluster [25]. Zhang et al. establishes a set-pair connection distance formula between samples and clustering centers, and manually adjusts parameters to divide different regions [29]. Fundamentally, the above algorithms cannot sufficiently account for both the inherent characteristics of every cluster and the positions of each object. This methodological limitation becomes particularly problematic when handling high-dimensional datasets, where optimal threshold determination should adaptively respond to data distribution characteristics rather than adhering to predetermined values. Therefore, it is essential to design a flexible strategy to adaptively construct three-way cluster representation in order to better recognize clusters with different distribution patterns [30].

To overcome the limitations of using fixed thresholds, Shah et al. [31] proposed an adaptive three-way clustering algorithm based on blurring and sharpening operations in image, namely, image blurring and sharpening inspired three-way clustering (BS3WC). The BS3WC converts a hard cluster into an image and defines cluster blur and cluster sharp operations, which are used to create three-way representation for clusters. In [32], the authors explored the role of blurring and sharpening in identifying the overlapping objects among the clusters and proposed a blur-sharp-based algorithm. The above two approaches both use operations of blurring and sharpening to concert 2W cluster to 3W cluster. The generations of 2W cluster are from conventional clustering algorithm. However, the samples in low-density areas from the original data will influence the clustering results. In order to improve the result of BS3WC, we propose three-way clustering based on Digital Image Processing, named 3WDIP in this paper. 3WDIP uses the concept of digital image processing and employ kernel density estimation function to represent the original data as gray values. By blurring and sharpening operations for the university, we delete these samples with low density and obtain samples with high density. 2W clusters are produced by using traditional clustering algorithm for samples with high density. Based on the assumption that objects with higher density are closer to core regions of clusters, we define the core region as the stable high-density pixels obtained by low-pass filtering (blurring). In the contrast, we construct the stable low-density pixels obtained by high-pass filtering (sharpening) as the trivial region. Blurring and sharpening operations respectively reveal the membership of stable objects in each cluster: definitely belonging and definitely not belonging. The remaining uncertain objects are naturally divided into the fringe region, and then the three-way cluster representation is completed. We validate the approach on various datasets including both synthetic and real world datasets using the benchmarks of the clustering indexes adjusted rand index (ARI), normalized mutual information (NMI) and adjusted mutual information (AMI), which demonstrate the effectiveness of our approach. The main contributions can be summarized as follows.

1) We propose a novel three-way clustering, called 3WDIP that utilizes the gray values of digital image processing to represent the different densities in data universe.
2) By blurring and sharpening operations for the university, the samples with low density are deleted and 2W clusters are produced by using traditional clustering algorithm for samples with high density.
3) To overcome the limitations of using fixed thresholds, an adaptive three-way cluster representation is devised that employs blurring and sharpening operations to reveal the intricate different distribution patterns among each cluster.

II. PRELIMINARY

A. THREE-WAY CLUSTERING

Assuming that a dataset $U = \{x_1, x_2, \cdots, x_n\}$ contains n objects, and $x_i \in \mathbb{R}^F$. $C = \{C_1, C_2, \cdots, C_k\}$ is the clustering result of the dataset U. The properties of traditional hard clustering for C_i are as follows:

1) $C_i \neq \emptyset, i = 1, 2, \ldots, k$
2) $\bigcup_{i=1}^{k} C_i = U$
3) $C_i \cap C_j = \emptyset, i \neq j$

Property.1 indicates that each cluster can not be empty, and property. (2)-(3) indicate that each object belongs to and

belongs to only one cluster. The above properties require that there must be the clear boundaries between the clustering results. However, there are often numerous uncertain objects in real-world scenarios. They could reduce the accuracy and effectiveness of the clustering results if they are arbitrarily assigned to a certain class.

To address the limitation, Yu introduces three-way decision into clustering and proposes three-way clustering for dealing with uncertainties [14], [27]. Different from hard clustering, three-way clustering divides the relationship between objects and clusters into: belonging to, may belong to, do not belong to. Therefore, three-way clustering of each cluster can be represented by a pair of sets: the core region (Co), the fringe region (Fr), and the trivial region (Tr):

$$C_i = \{Co(C_i), Fr(C_i)\},\ Tr(C_i) = U - (Co(C_i) \bigcup Fr(C_i)). \tag{1}$$

Three sets are used to represent a class, which fully considers the objects with insufficient information that cannot determine the cluster attribution. Meanwhile, this method can efficiently reduce the risk of decision-making and be widely used in attribute reduction, uncertainty analysis.

B. KERNEL DENSITY ESTIMATION

Kernel Density Estimation (KDE) is a non-parametric method for estimating the probability density function of random variables [34], [35]. KDE calculates the probability distribution of the dataset by applying a smooth kernel function to each observation around the objects. This method does not need to assume the distribution of data in advance, and can generate a smooth density estimation curve. The density of object x_i can be defined as:

$$\rho_i = \sum_{x_j \in U} K\left(\frac{\|x_j - x_i\|}{h}\right), \tag{2}$$

where h is the bandwidth parameter, which determines the width of the kernel function and affects the smoothness of KDE. The choice of bandwidth has a great impact on the effect of KDE. Too small bandwidth leads to over fitting, resulting in many wavelet peaks; Too large bandwidth leads to under fitting, making the estimation result too smooth and losing details of datasets. K is a non-negative monotone decreasing kernel function. Generally, the kernel function K usually selects functions that conform to specific properties, such as Gaussian kernel (normal distribution function) which can be represented as:

$$K(x_i, x_j) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\|x_j - x_i\|^2}. \tag{3}$$

Kernel density estimation can provide smooth and continuous density estimation, which effectively reveals the potential structure and distribution pattern of data. Thus, it is widely used in the field of cluster analysis.

C. BLURRING AND SHARPENING IN IMAGE PROCESSING

In image processing, blurring and sharpening are two commonly used operations to change the visual effect and quality of images.

Blurring operation, i.e. low-pass spatial filter, is used to reduce the interference of noise and ensure the integrity of image information, also known as image smoothing. There are two kinds of blurring operations commonly used: spatial method and frequency method. The spatial method uses modular operation and different fuzzy operators to filter the target pixels, so as to extract the characteristics of the neighborhood pixels. The frequency method uses orthogonal transformation and high-frequency characteristics of objects to filter. In this paper, we employ the spatial method (mean filtering LP) to blur the cluster, which can be denoted as:

$$Lp[f(x_i)] = \frac{1}{N_s} \sum_{x_j \in S(x_i)} f(x_j), \tag{4}$$

where $f(x_i)$ is the density estimation of object x_i, $S(x_i) = \{x_1, \cdots, x_j\}$ is the neighborhood set of object x_i and $|S| = N_s$

Sharpening operation, i.e. high-pass spatial filter, emphasizes the high-frequency components (details and edges) in the image by defining differential operators. By subtracting the blurred image from the original image, we have information of the edges when added to the original image produces a sharp image. The sharpening operation for the object x_i is defined as:

$$Sp[f(x_i)] = f(x_i) + \alpha(f(x_i) - Lp[f(x_i)]), \tag{5}$$

where α is a scaling factor for controlling the sharpening effect. Mapping this operation to the clusters, sharpening operation enhances the high-density edge regions of the cluster. In addition, sharpening can also weaken areas with low data density and reduce the density of outliers (low-frequency signals).

III. METHOD

A. BINARY GRAY SCALE REPRESENTATION BASED ON KERNEL DENSITY ESTIMATION

In order to represent the original data as a gray scale image, 3WDIP samples and quantizes the features of input dataset. Among them, sampling determines the resolution of gray scale image, which is the number of image pixels obtained by converting the cluster. Quantization is the process of assigning gray scale values to each pixel based on kernel density estimation. Immediately, each cluster of the dataset can be converted into gray scale images through sampling and quantization. Kernel density estimation is a widely used density measurement method, defined as Eq.(1). In this paper, we employ inverse multiquadric kernel function to estimate the density of each object, which can be represented as:

$$K(x_i, x_j) = \frac{1}{\sqrt{\|x_i - x_j\|^2 + c^2}} \tag{6}$$

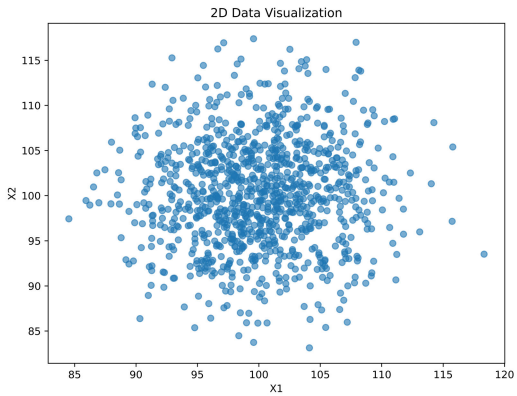


FIGURE 2. 2D data visualization.

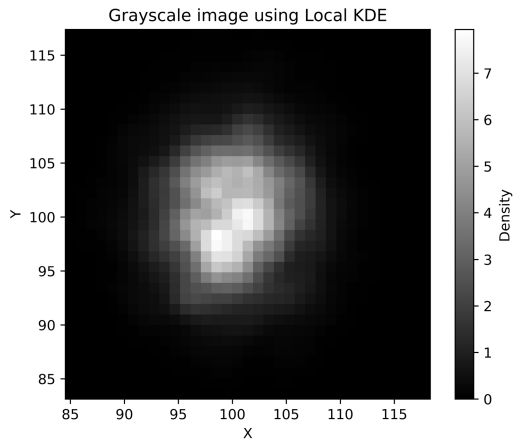


FIGURE 3. Gray scale image.

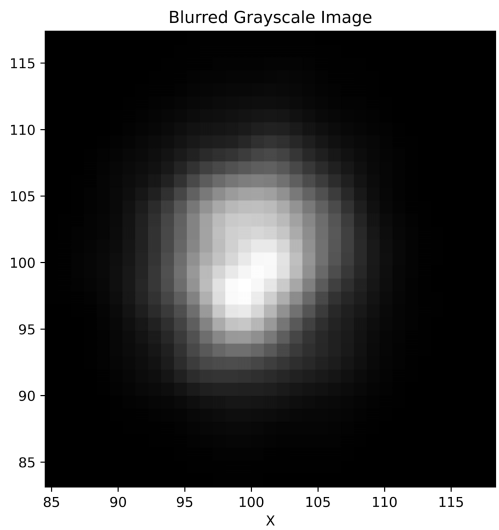


FIGURE 4. Blurred gray scale image.

where c is a fuzzy coefficient that can balance the density between objects with different distributions. Inverse multiquadric kernel function is a highly localized function with a bell shaped curve. Furthermore, this function has two major advantages: fuzzy coefficient c is set to the diagonal length of the pixel box, which can adapt to different data distributions without the need for user specified coefficients; Fuzzy coefficient c helps to avoid grayscale imbalance in representing objects. It ensures that objects in low-density areas do not get assigned overly low values, while objects in high-density areas do not receive excessively high values. Therefore, the fuzzy coefficient c is defined as:

$$c = \sqrt{|x_{1(max)} - x_{1(min)}|^2 + \cdots + |x_{F(max)} - x_{F(min)}|^2}. \quad (7)$$

According to inverse multiquadric kernel function, the density estimate of each object can be obtained. When the density of the object is low, the object is assigned a smaller gray scale value (the pixel color is closer to black). In contrast, the object is assigned a larger gray scale value (the pixel color is closer to white) while its density is high.

We sample neighboring set of each object and obtain density estimates based on the kernel function, and then quantize the density values into $(0-1)$ interval. The selected neighborhood set is generally 1/10 of the dataset size. Given a two-dimensional dataset and its distribution characteristics are a normal distribution with a mean of 100 and a variance of 5. After the above operation, the gray scale image can be obtained as shown in Fig. 2 and Fig. 3.

As shown in Fig. 2 and Fig. 3, we have the observations that high density objects are closer to the core region of three-way representation and surrounding objects are more likely to belong to the fringe region. Based on gray scale value, the next step is to design an adaptive method to obtain three-way cluster representation.

B. BLURRING AND SHARPENING PROCESSING

Blurring Let the low-pass filter be $Lp[\cdot]$, and the set $S(x_i) = \{x_1, \cdots, x_j\}$ contains the neighborhood objects of x_i. We employ the mean filtering to obtain the blurred gray scale image, as shown in Fig. 4.

The specific operation of blurring is to return the gray scale mean of pixels within the radius r of adjacent pixels $(f(x_j),\ x_j \in S(x_i))$ to the pixel point $f(x_i)$. In clustering, the fuzzy process represents that the target object aggregates the density values of neighboring samples, which can perceive local information. In addition, while the density of neighborhood objects is high or low, the blurring operation does not affect the density of the target object, which reserve the distribution of stable objects during clustering. For onjects with uneven density of neighbors, the fuzzy operation updates the targat object to the mean of local densities.

Sharpening Let the high-pass filter be $Sp[\cdot]$, and we employ the strategy of difference supplement to design a novel high-pass filter, as shown in Fig. 5. Specifically, we calculate the difference between the original image and the blurred image, and add it to the original image to get the sharpened image. The sharpening operation avoids the boundary gray value prominent and does not expand the region of each cluster. After sharpening, the high-density region is expanded, so as to effectively reduce the impact of the region with low gray value on the clustering results.

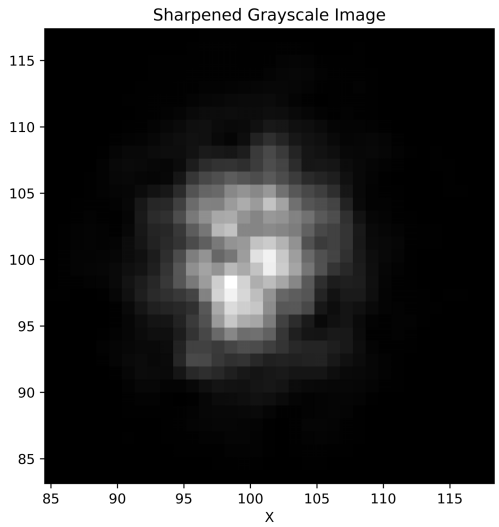


FIGURE 5. Sharpened gray scale image.

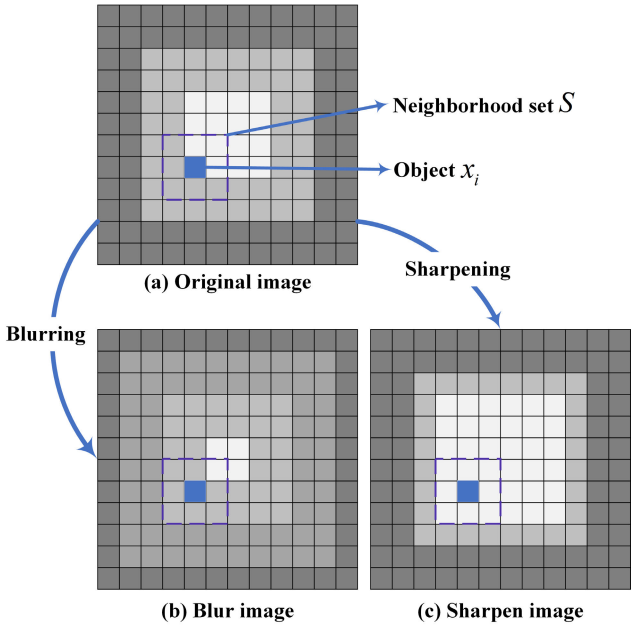


FIGURE 6. Blurring and sharpening processing.

C. CONSTRUCTING THREE-WAY CLUSTER REPRESENTATION

In order to better characterize the uncertainty of data, we use fuzziness and sharpening to adaptively obtain the core and boundary regions of each cluster. Summarizing the blur and sharpen operations, we clearly show their effects in Fig. 6. After the fuzzy operation, the high gray area is shrunk to get the more stable area, which corresponds to the core region of three-way clustering. Therefore, we define the remaining stable region as the core region (Co) of this cluster C_i, which can be denoted as:

$$Co(C_i) = \{x_i \in U \mid x_i \in C_i \wedge Lp[f(x_i)] = 1\}, \tag{8}$$

where $Lp[f(x_i)] = 1$ indicates that the gray value of the target is 1 before and after the blurring. In this case, the neighborhood of the target object is all 1, which corresponds to the stable high-density region. After sharpening, the high-density region in the image is expanded, which means that the expanded part is closely related to the core region. Intuitively, we define this region as the fringe region (Fr), which is expressed as:

$$Fr(C_i) = \{x_i \in U \mid x_i \in C_i \wedge Lp[f(x_i)] \neq 1 \wedge Sp[f(x_i)] = 1\}, \tag{9}$$

Finally, the remaining part is always a low-density area, indicating that its gray value is too small and too distant from the core region. Therefore, we define this region as a trivial region (Tr), which is represented as:

$$Tr(C_i) = \{x_i \in U \mid x_i \in C_i \wedge Lp[f(x_i)] \neq 1 \wedge Sp[f(x_i)] \neq 1\}. \tag{10}$$

Through the above operations, we can obtain the core region (Co), fringe region (Fr) and trivial region (Tr) of each cluster. Among them, the objects in the fringe region are closely related to the core region, while the trivial region contains data with weak relationships with the cluster, such as noise and outliers. Therefore, it is believed that trivial regions are not part of their cluster, and then the three-way clustering results of the dataset U are obtained as:

$$C = \{(Co(C_1), Fr(C_1)), \cdots, (Co(C_k), Fr(C_k))\}. \tag{11}$$

D. PROCESS OF 3WDIP

The basic idea of 3WDIP is to apply blurring and sharpening from digital image processing to extract distribution characteristics from the dataset. Specifically, we perform kernel density estimation on the original data to generate visualized gray scale images for two-dimensional data and obtain gray scale values for high-dimensional data. Firstly, we use blurring and sharpening operations on each cluster to extract stable high-density regions and stable low-density regions, respectively. Secondly, we retain the objects with higher gray scale values after sharpening and remove low-density areas from the original data. Then, traditional clustering algorithms are used to obtain different clusters for data with higher gray scale values, which can effectively reduce the impact of areas with lower gray scale values on the clustering results. In order to better characterize the uncertainty of data, we use blurring and sharpening of each cluster to obtain the core region (Co) and fringe region (Fr), thereby obtaining the three-way clustering results.

Algorithm 1 presents the process of Three-way clustering based on Digital Image Processing (3WDIP).

IV. EXPERIMENTS

A. DATASETS

To verify the effectiveness of 3WDIP, we select 12 UCI datasets for experimentation. The dataset details are shown in Table 1, with some data added with noise.

Algorithm 1 3WDIP

Input: Original data U; number of cluster k;

Output: $C = \{(Co(C_1), Fr(C_1)), \cdots, (Co(C_k), Fr(C_k))\}$;

Sample and obtain the fuzzy coefficient c in the kernel function;

Apply kernel density estimation to obtain original gray scale values $f(U)$;

Employ $f(U)$ to obtain blurred gray scale $Lp[f(U)]$;

Combining $f(U)$ and $Lp[f(U)]$ to obtain sharpened gray scale $Sp[f(U)]$

Cluster the sharpened extracted region to get the original results $C = \{C_1, \cdots, C_k\}$;

Divide into core region (Co) and fringe region (Fr) separately.

Obtain the three-way cluster representation $C = \{(Co(C_1), Fr(C_1)), \cdots, (Co(C_k), Fr(C_k))\}$.

TABLE 1. Data details.

Datasets	Instance	Features	Classes
Iris	150	4	3
Libras	360	90	15
Seeds	210	7	3
Segmentation	2310	19	6
Wine	178	13	3
Forest	523	27	4
Glass	214	9	6
Breast	106	9	6
Card	2126	21	10
Page	5473	10	5
Thyroid	215	5	3
D10	1426	2	4
Pathbased	300	2	3

B. EVALUATION METRICS

Adjusted Rand Index (ARI) is an improved Rand index that normalizes random clusters to make the evaluation of clustering effects more fair and intuitive. It is widely used in clustering evaluation scenarios with real labels, especially for handling imbalanced data, which is defined as:

$$\mathrm{ARI} = \frac{\mathrm{RI} - \mathbb{E}[\mathrm{RI}]}{\max(\mathrm{RI}) - \mathbb{E}[\mathrm{RI}]}, \tag{12}$$

where RI is rand index, and $\mathbb{E}$ is expected value of Rand coefficient.

Standardized Mutual Information (NMI) is a commonly used and effective external clustering evaluation metric that can measure the degree of information sharing between clustering results and real labels. It has advantages such as standardization, symmetry, and robustness to cluster imbalance and can be represented as follow:

$$\mathrm{NMI(Y, Y')} = \frac{\mathrm{MI(Y, Y')}}{\sqrt{\mathrm{YY'}}} \tag{13}$$

where Y is the ground truth and Y' is the predict result of clustering. Adjustment of Mutual Information (AMI) is an external clustering evaluation metric based on information theory, which adjusts the mutual information in random situations to make the expected value of the random clustering result zero. Compared to standardized mutual information (NMI), AMI can more accurately reflect clustering effects and is particularly suitable for handling imbalanced data or scenarios with clusters of different sizes.

$$\mathrm{AMI(Y, Y')} = \frac{\mathrm{MI(Y, Y')} - \mathbb{E}\{\mathrm{MI(Y, Y')}\}}{\max(\mathrm{H(Y)}, \mathrm{H(Y')}) - \mathbb{E}\{\mathrm{MI(Y, Y')}\}} \tag{14}$$

C. EXPERIMENTAL RESULTS

1) COMPARISON OF CLUSTERING RESULTS

Firstly, we perform kernel density estimation on original data to obtain the gray scale values on each object, and then perform blurring and sharpening operations to extract the sharpened high-density regions. Then, we use k-means to obtain hard clustering results, and then divide each cluster into core region (Co) and fringe region (Fr). To demonstrate the effectiveness of the proposed algorithm, Table 2 shows the experimental results of 3WDIP embedding K-means.

A comparative analysis of mean values across the evaluated datasets was conducted to generate the performance indicator comparison shown in Fig. 7. The negative values in this visualization denote a reduction in clustering accuracy following the application of the 3WDIP methodology, suggesting potential trade-offs between data preprocessing intensity and algorithm effectiveness.

According to the Table 2 and Fig. 7, we have the following observations. By comparing the NMI standard mutual information, it was found that when the 3WDIP algorithm is applied to K-means, the numerical amplification of each item is relatively stable. After applying the 3WDIP algorithm, the ARI and NMI values of the dataset Page are 0.2856 and 0.4939, while the two values obtained by running only K-means are 0.0601 and 0.1166. This improvement is attributed to the elimination of noise and outlier interference in sharpening extraction, clustering only the core and fringe regions, and clarifying the boundaries between each cluster.

In order to further verify the effectiveness of 3WDIP, the high-density regions extracted from the above datasets are sharpened, and the hard clustering results are obtained by DPC clustering. Then core regions and fringe regions of each class are obtained by blurring and sharpening each class cluster. Figs. 8,9,10 show the comparisons of clustering results obtained by embedding DPC with 3WDIP and those obtained by directly using DPC on ARI, NMI and AMI.

The observations show that the average increase of 3WDIP with DPC is greater than with K-means, mainly due to the blur and sharpen processing. In DPC, the gray value corresponds

TABLE 2. Clustering results.

Data sets	3WDIP-kmeans			k-means		
	ARI	AMI	NMI	ARI	AMI	NMI
Iris	**0.770±0.017**	**0.818±0.032**	**0.823±0.031**	0.719±0.052	0.744±0.051	0.747±0.051
Libras	**0.417±0.006**	**0.602±0.013**	**0.655±0.010**	0.285±0.039	0.510±0.041	0.568±0.034
Seeds	**0.762±0.026**	**0.781±0.007**	**0.785±0.007**	0.711±0.014	0.696±0.011	0.699±0.011
Segmentation	**0.465±0.051**	**0.644±0.053**	**0.646±0.053**	0.374±0.031	0.532±0.038	0.534±0.039
Wine	**0.415±0.003**	**0.555±0.003**	**0.568±0.001**	0.366±0.005	0.420±0.002	0.426±0.002
Forest	**0.493±0.090**	**0.593±0.054**	**0.597±0.053**	0.383±0.113	0.467±0.081	0.470±0.080
Glass	**0.241±0.032**	**0.417±0.067**	**0.449±0.065**	0.157±0.023	0.266±0.025	0.297±0.024
Breast	**0.317±0.017**	**0.504±0.026**	**0.556±0.024**	0.322±0.083	0.465±0.018	0.507±0.019
Card	**0.238±0.028**	**0.416±0.025**	**0.422±0.025**	0.162±0.019	0.319±0.025	0.325±0.025
Page	**0.255±0.031**	**0.478±0.027**	**0.479±0.027**	0.054±0.011	0.100±0.015	0.101±0.015
Thyroid	**0.420±0.023**	**0.573±0.042**	**0.581±0.083**	0.579±0.065	0.488±0.035	0.488±0.035
Pathbased	**0.664±0.038**	**0.744±0.054**	**0.747±0.034**	0.464±0.024	0.545±0.022	0.548±0.020

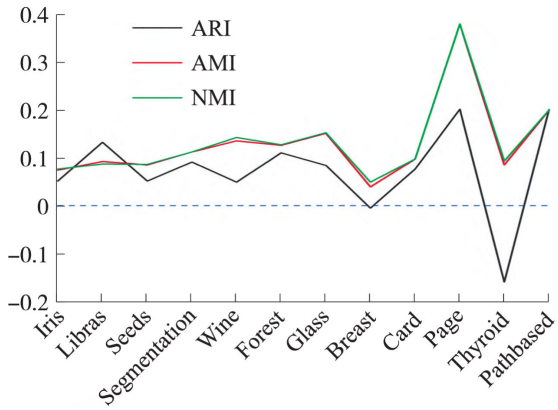


FIGURE 7. Comparison of clustering indicators.

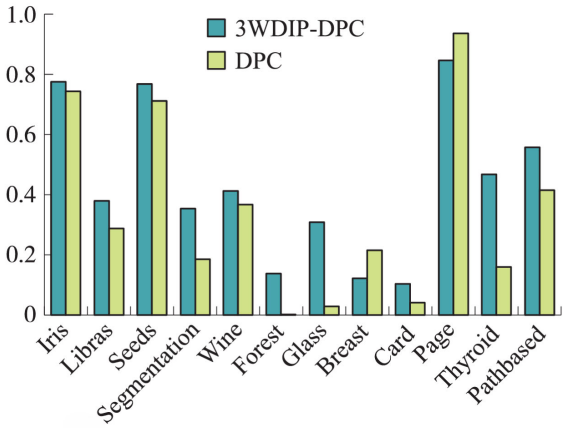


FIGURE 8. ARI obtained by embedding DPC.

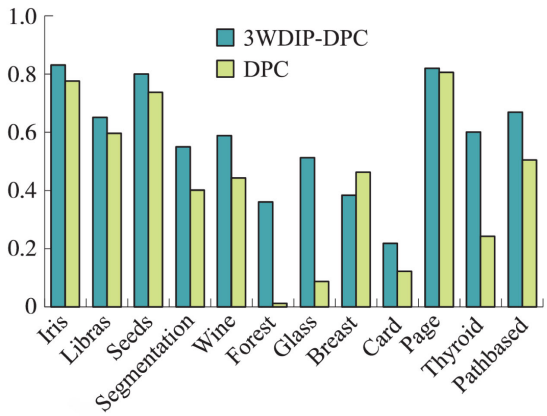


FIGURE 9. NMI obtained by embedding DPC.

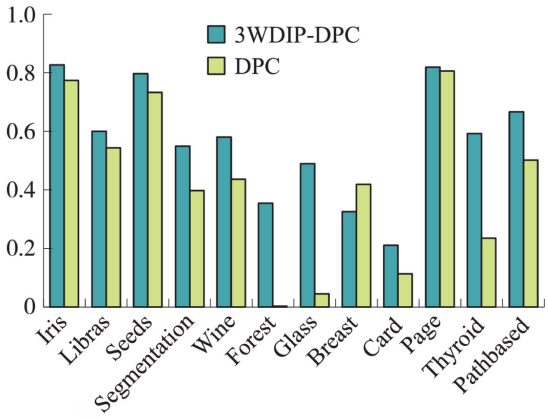
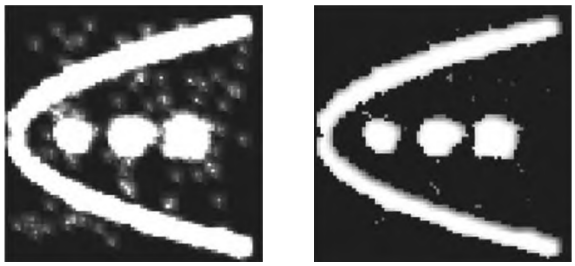


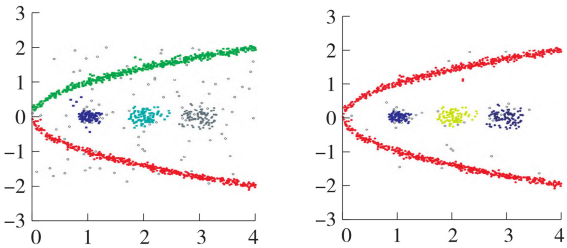
FIGURE 10. AMI obtained by embedding DPC.

to the density, and after the blur and sharpen steps, the gray values in the boundary and core regions of each cluster are enhanced. This process minimizes the correlation between clusters and their surrounding areas, thus improving intra-cluster cohesion.



(a) Original data (b) Sharpened region

FIGURE 11. Comparison of gray scale images using 3WDIP.



(a) Dbscan (b) 3WDIP+Dbscan

FIGURE 12. Comparison of clustering results in the core region.

However, the failure to improve the performance on the Beast dataset is due to the uneven distribution of data within clusters. The presence of saddle points in the data causes 3WDIP to reduce the gray values around these points, which negatively impacts DPC's effectiveness. To sum up, the 3WDIP can better process the data characteristics and be applied to different clustering methods. Its three-way clustering results can solve the uncertain clustering problems in reality.

2) VISUALIZATION ANALYSIS

To further demonstrate the effectiveness of the 3WDIP algorithm, we selected D10 data to display its gray scale image and the clustering results of Dbscan and 3WDIP+DBscan, as shown in Fig. 11 and Fig. 12.

It is clear from Fig. 11 that the sharpening extraction of 3WDIP removes most of the outliers, making the contours between clusters clearer. In addition, in Fig. 12, the Dbscan algorithm incorrectly divides the arc-cluster into two parts. This is because there are sparse samples in the middle of the arc cluster, which reduces the density. For the 3WDIP algorithm, the fuzzy processing makes the samples perceive the local distribution, so as to update their own density and achieve better clustering results.

V. CONCLUSION

This paper combines the concepts of blurring and sharpening in digital image processing with three-way clustering to propose three-way clustering based on digital image processing (3WDIP). The 3WDIP methodology integrates digital image processing principles with kernel density estimation (KDE) to transform raw data points into gray scale intensity mappings. Through successive blurring and sharpening operations applied to the data distribution, this approach effectively eliminates low-density outliers while preserving high-density core regions. Subsequently, conventional clustering algorithms are implemented on the refined high-density subset, yielding 2W distinct clusters that represent essential data patterns. For each cluster, the blurring operator is used to identify the core region, while the sharpening operator defines the cluster's fringe region, thereby providing a three-way representation of each cluster. Experimental results show that the algorithm enhances the display of core and fringe region, improving the separation between clusters and stabilizing clustering performance.

However, the algorithm has areas that require improvement. For instance, it weakens regions within clusters where data dispersion creates saddle points, which can lead to increased dispersion within clusters. Additionally, another area for future research is the optimization of data feature extraction to further enhance the algorithm.

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